



# An Ontology-Guided Approach for Enhanced Documentation of Cave Microbial Culture Collections

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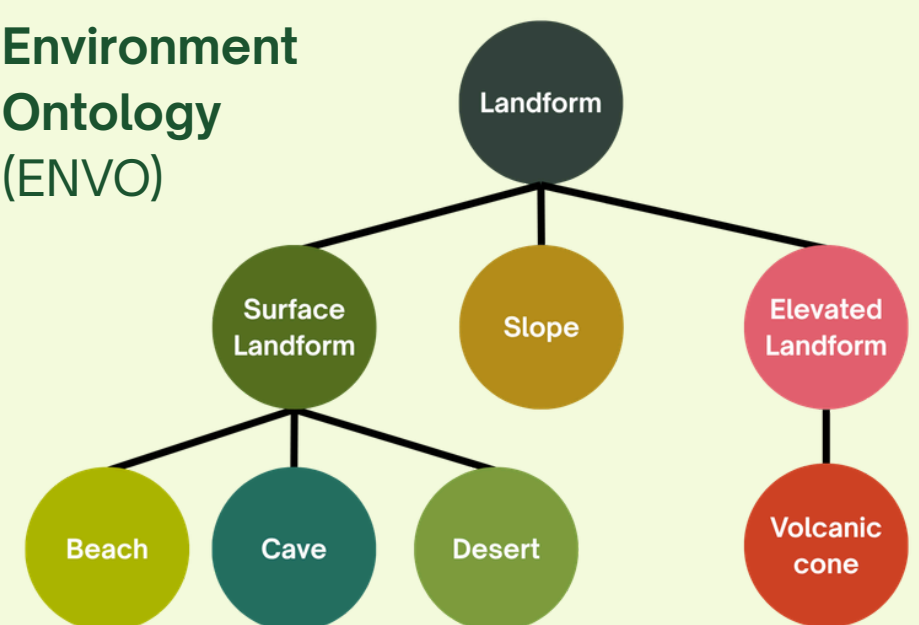
## 1 Introduction

Free-text annotations in culture collection systems lead to **inconsistent** descriptions of **identical** concepts, making cave microbial data difficult to interpret and compare.

low temperature == cold ?

**Ontologies** bring a unique solution by providing **standardized terms** with defined relationships, promoting accuracy in data entry.

### Environment Ontology (ENVO)



## 2 Objectives

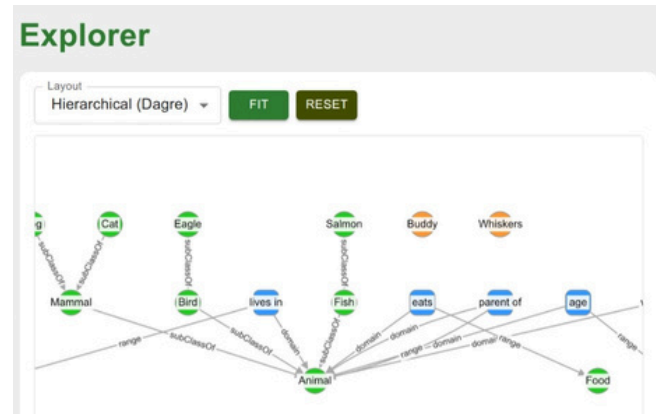
Escherichia coli  
<http://purl.bioontology.org/ontology/NCBITAXON/562>

Balantoides coli  
<http://purl.bioontology.org/ontology/NCBITAXON/71585>

Entamoeba coli  
<http://purl.bioontology.org/ontology/NCBITAXON/110766>

Aedoeadaptatus coli  
<http://purl.bioontology.org/ontology/NCBITAXON/2058292>

- to integrate **ontology-based suggestions** on free-text annotations to support more consistent data entry;



- to develop a Graphical User Interface (GUI) for an **Ontology Lookup Service (OLS)** and an **Ontology Explorer Service (OES)** to help users explore relevant ontology classes and their relationships; and

- to evaluate the usability of the system to be developed using the **System Usability Scale (SUS)**.

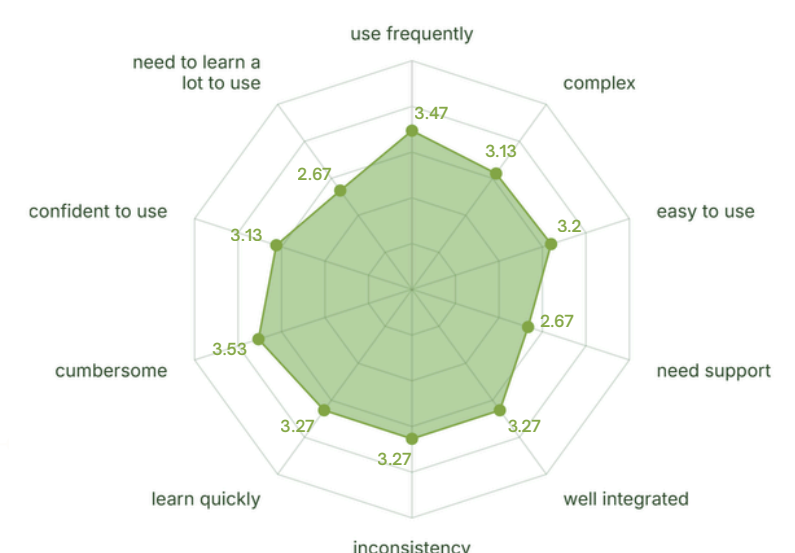
## 3 Methodology

The Caventology web application was developed with **React.js** for the user interface, **Material UI (MUI)** for interface components and styling, **Django** for the server-side application, and **MySQL** for database management. The OES uses **N3.js** for parsing RDF ontology files and **Cytoscape.js** for visualization. On the other hand, the OLS and ontology-based suggestions uses the **BioPortal Search API**.



## 4 Results

- 15** respondents evaluated the system using **SUS**.
- Researchers appreciate the **tight integration** of the application's features.
- The system received a mean SUS score of **79**, indicating **good** overall usability.



## 6 Conclusion

By integrating ontology-based suggestions along with OLS and OES tools in the cave culture collection system, **Caventology** enhances the consistency and semantic quality of cave microbial documentation.



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